

Unit 08: Network layer – routing

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Objectives

- what is Unicast Routing.
- three major protocols for unicast routing.
- classification of Routing Algorithms.

Introduction

At the network layer, the Internet Protocol (IP) serves as the primary protocol. IP is a best-effort distribution protocol, but it lacks certain functionality including flow control and error control. It's a logical addressing-based host-to-host protocol. Other protocols are needed to make IP more open to certain specifications of today's internetworking. To construct a mapping between physical and logical addresses, protocols are used. Logical addresses are used in IP packets. However, these packets must be encapsulated in a frame, which requires physical addresses (node-to-node). We'll see the ARP, or Address Resolution Protocol, is designed specifically for this purpose. We sometimes need to reverse map a physical address to a logical address. Since the Internet Protocol lacks flow and error management, another protocol, ICMP, was created to provide warnings. It records network or destination host congestion and other forms of errors.

9.1 Unicast

Unicast routing refers to propagation from a single source to a single receiver, often known as point-to-point correspondence between sender and receiver. TCP and HTTP are two examples of unicast protocols. TCP stands for transmission control protocol, which is the most widely used unicast protocol. It is a connection-oriented protocol that relies on the recipient site's recognition. The hypertext transmission protocol (HTTP) stands for hypertext transfer protocol. It is a collaboration protocol that is based on object-oriented principles. Intradomain routing protocol, intranetwork routing protocol, and intranet routing protocol are two types of routing protocols as shown in Figure 1. Intradomain routing protocols are used to route packets within a given area, such as within an institutional network for e-mail or Web browsing. An interdomain routing protocol, on the other hand, is a method for routing packets through domain networks.

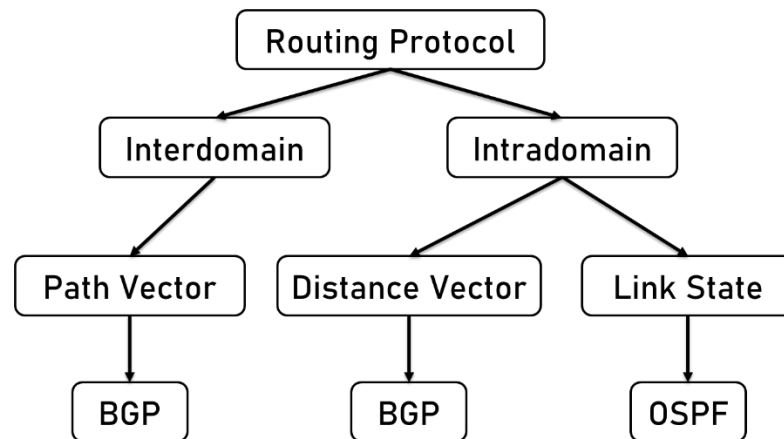


Figure 1 Types of Routing Protocol

For unicast routing, we have three main protocols: distance vector routing, connection state routing, and route vector routing as shown in Figure 2.

9.2 Distance Vector Routing

In distance vector routing, the least cost for any two nodes is the path with the shortest distance, and each node keeps a table of the shortest distances to each other. The term "distance vector" refers to how routes are marketed using two characteristics.

Distance: This metric determines how long it is to the destination network and is dependent on hop count, cost, bandwidth, distance, and other factors.

Vector: The vector specifies the next-hop router's or exit interface's path in order to reach the destination.

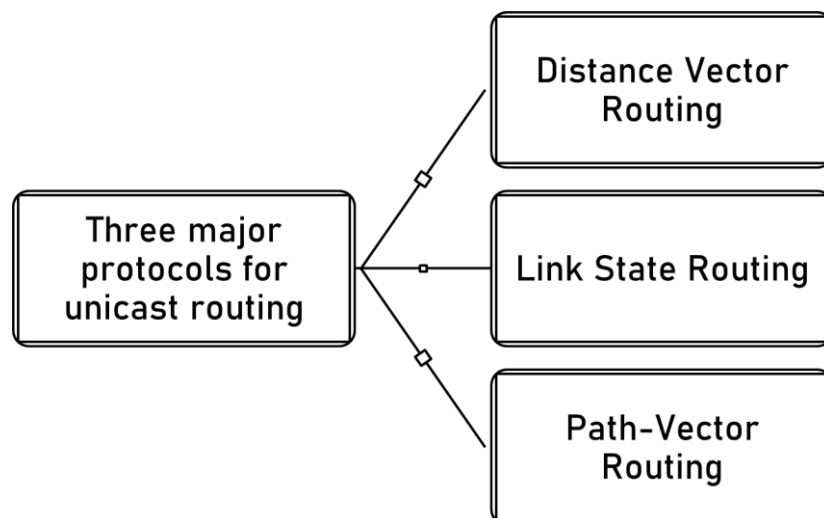


Figure 2 Three major protocols for unicast routing

Example: We present a system of five nodes and the tables that go with them in Figure 3.

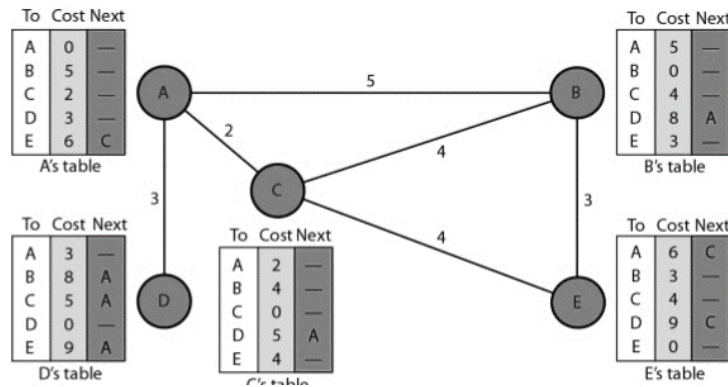


Figure 3 Distance Vector Routing Tables

The table for node A illustrates how to get from this node to every other node.

Our lowest cost to enter node E, for example, is 6. C is a stop along the way.

Initialization

Figure 3 shows that the tables are stable; each node knows how to access every other node and how much it will cost. However, this is not the case in the beginning. Only the distance between each node and its immediate neighbours, all that are closely bound to it, can be known. So, for the time being, we'll presume that each node will send a message to its immediate neighbours and calculate the distance between them.

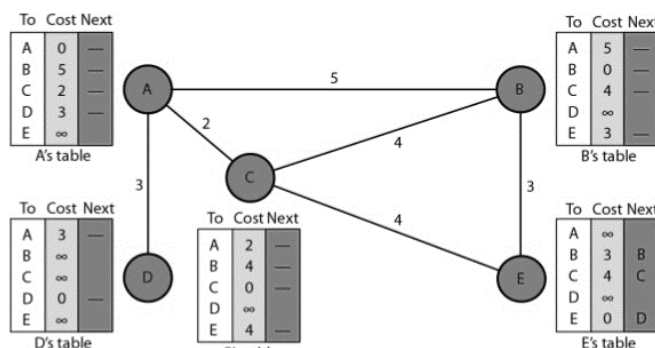


Figure 4 Initialization of Tables in Distance Vector Routing Protocols

The original tables for each node are seen in Figure 4. Any entry that isn't a neighbour is given an infinite distance (unreachable).

Sharing

The exchange of information between neighbors is at the heart of distance vector routing. While node A is unaware of node E, node C is. As a result, if node C shares its routing table with node A, node A will still figure out how to get to node E. Node C, on the other hand, has no idea how to get to node D, but node A does. If node A and node C share a routing table, node C will also know how to access node D. In other words, as immediate neighbors, nodes A and C will boost their routing tables by assisting one another. There's just one problem. How much of the table does each neighbour share? The table of a neighbour is unknown to a node. The best answer for each node is to give its entire table to its neighbour and let him determine the parts to keep and which to delete.

The neighbour, on the other hand, does not benefit from the third column of a table (next stop). This column must be replaced with the sender's name when the neighbour collects a table. The sender of the table is the next node if all of the rows can be included. As a result, a node can only give the first two columns of its table to all of its neighbours. To put it another way, sharing here just refers to the first two columns.

Updating

A node must change its routing table when it receives a two-column table from a neighbour. There are three approaches to updating:

1. For each value in the second column, the receiving node must apply the cost between itself and the transmitting node.
2. If the receiving node uses information from another row, it must apply the name of the transmitting node as the third column of each row. The next node in the path is the sending node.
3. Each row of the receiving node's old table must be compared to the corresponding row of the updated version of the received table.
 - a. Where the next-node entry differs, the receiving node selects the row with the lowest rate. The old tie is maintained if there is one.
 - b. The receiving node selects the new row if the next-node entry is the same. Assume node C previously advertised a distance-based path to node X. Assume there is no longer a path between C and X; node C now advertises this route with an infinite radius. About the fact that its old entry is smaller, Node A must not disregard this value. The old road is no longer in use. The current path is infinite in duration.

Node A changes its routing table after obtaining the partial table from node C, as seen in Figure 5. When Do You Share?

When does a node transmit its partial routing table (which only has two columns) to all of its immediate neighbours? The table is submitted both on a regular basis and when the table changes.

Periodic Update: Every 30 seconds, a node sends its routing table in a periodic update. The length of time is determined by the protocol that uses distance vector routing.

Triggered Update: When a node's two-column routing table changes, it sends a triggered update to its neighbours. This is referred to as an activated upgrade.

The following factors can influence the outcome.

1. A node receives a table from a peer and updates it, resulting in modifications in its own table.
2. A node senses a defect in one of the adjacent connections, resulting in a drop in distance to infinity.

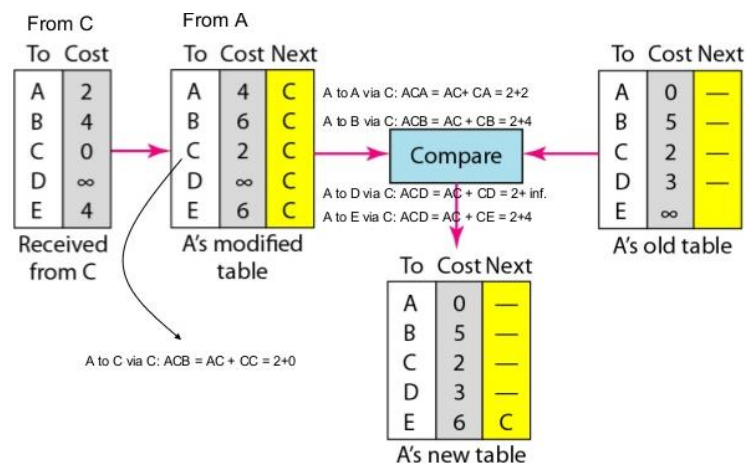


Figure 5 Updating in Distance Vector Routing Table

9.3 Link State Routing

The second family of routing protocols is connection state routing. Link-state routing uses link-state routers to exchange messages that allow each router to learn the entire network topology, while distance vector routers use a distributed algorithm to compute their routing tables. Each router will then compute its routing table using a shortest path calculation based on the learned topology.

Features of link state routing protocols

Link State Packet: A small packet that includes routing information is known as a connection state packet.

Link State Database – A database containing data collected from link state packets.

Shortest path first algorithm (Dijkstra algorithm) – A database estimation produces the shortest path.

Routing Table: A list of known routes and interfaces is called a routing table.

Calculation of shortest path

To find shortest path, each node need to run the famous **Dijkstra algorithm**. This famous algorithm uses the following steps:

Step-1: The node is selected as the tree's root node, resulting in a tree with just one node. Set the cumulative cost of each node to a value depending on the information in the Connection State Database.

Step-2: The node now chooses the node that is closest to the root of all the nodes not in the tree-like structure and connects it to the tree.

The tree's outline is transformed.

Step 3: After this node has been connected to the stack, all non-tree nodes' costs must be modified so the paths could have shifted.

Step-4: The node repeats Steps 2 and 3 until all of the nodes in the tree have been inserted.

In contrast to Distance Vector protocols, Link State protocols have:

1. It necessitates a significant amount of memory.
2. Shortest route computations necessitate a large number of CPU cycles.
3. If a network uses a small amount of bandwidth, it responds easily to topology changes.
4. To shape connection state packets, all objects in the database must be sent to neighbours.
5. In the topology, all neighbours must be trusted.
6. To prevent unwanted adjacency and complications, authentication methods may be used.
7. In link state routing, split horizon strategies are not feasible.

Open shortest path first (OSPF) routing protocol –

1. OSPF (Open Shortest Path First) is a unicast routing protocol developed by an Internet Engineering Task Force working group (IETF).
2. It's a protocol for routing within a domain.
3. It's a free and transparent protocol.
4. It's akin to the Routing Information Protocol (RIP) (RIP).
5. OSPF is a classless routing protocol, meaning it uses the subnet of each path it recognises in its notifications, allowing for variable-length subnet masks. An IP network can be divided into several subnets of different sizes using variable-length subnet masks. This gives network operators more consistency in network setup.
6. OSPF is implemented as a network layer application that utilises the facilities offered by the Internet Protocol IP datagram that transports OSPF messages. OSPF sets the protocol field value to 89.
7. The OSPF algorithm, also known as the Dijkstra algorithm, is based on the SPF algorithm.
8. Version 1 and version 2 of OSPF are available. Version 2 is the most widely used.

OSPF Messages – The OSPF protocol is a complicated one. It employs five distinct message forms. The following are some of them:

1. Hello messages (Form 1)
2. Message from the database (Type 2)
3. Message with a link-state request (Type 3)
4. Message about a link-state update (Type 4)

5. Message of Link-State Recognition (Type 5)

9.4 Path Vector routing

The cost of reaching a given destination is not used by a route vector protocol to decide whether or not each path open is loop free. Instead, route vector protocols analyse the path to the destination to determine if it is loop-free.

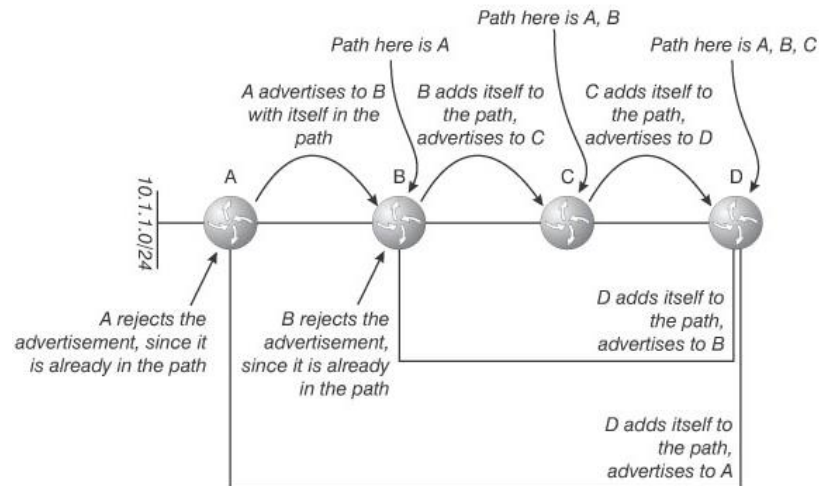


Figure 6 Path Vector Routing

By logging each hop the routing advertising traverses through the network, a route vector protocol ensures loop-free routes through the network. In this case, router A informs router B that the 10.1.1.0/24 network is reachable. When router B receives this data, it joins the path and broadcasts it to router C. Router C joins the road and informs router D that the 10.1.1.0/24 network can be reached in this direction. As Router D receives the route advertising, it joins the road as well. When router D tries to advertise to router A that it will hit 10.1.1.0/24, router A rejects the advertising because the corresponding direction vector in the advertisement suggests that router A is already in the path. When a router encounters a message in which it is already a part of the path, it rejects it because accepting the request will result in a routing information loop.

9.5 Border Gateway Protocol(BGP)

Rather than considering a single router as a single point in the path to any given destination, BGP applies the path vector principle on a broader scale. Each autonomous system is treated as a single point on the road to any given destination by BGP as shown in Figure 7.

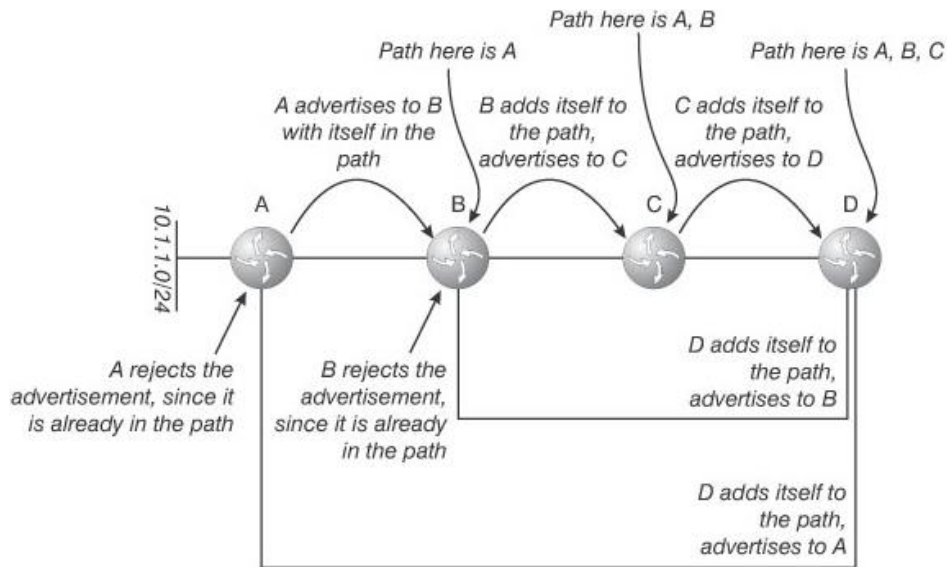


Figure 7 Path vector over a set of autonomous systems

This case is similar to Figure 6, except that instead of a single router, each autonomous device is treated as a point along the road. The network 10.1.1.0/24, also known as a prefix, is marketed along with a list of autonomous systems over which the upgrade has passed; this list of autonomous systems is referred to as the AS Route. AS 65100 creates the prefix 10.1.1.0/24 and advertises it to AS 65200, connecting itself to the AS Route. AS 65200 is added to the AS Route, and the prefix is advertised to AS 65300.

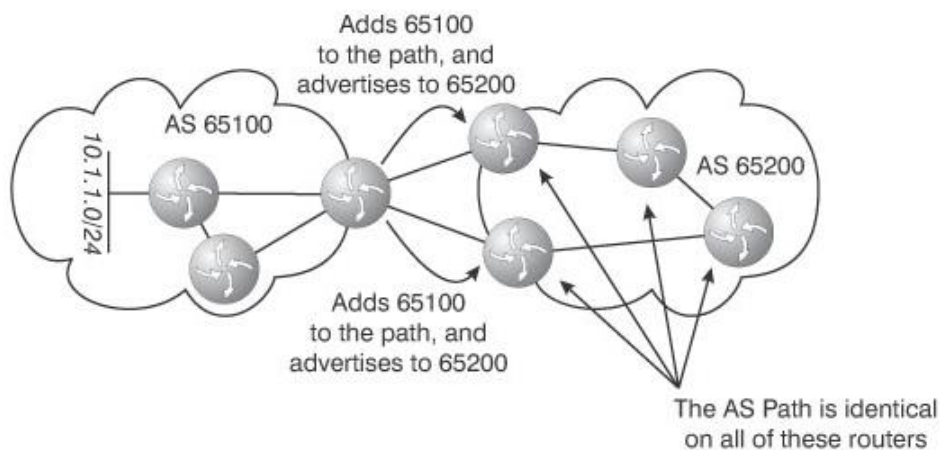


Figure 8 BGP routing with an AS

When AS 65300 advertises the prefix 10.1.1.0/24 to AS 65100, it is refused because the 65100 sees that its local AS is already in the AS Path, and accepting the route will result in a routing information loop as shown in Figure 8. The main reason BGP considers an entire autonomous device as a single hop in the AS Path is to mask the AS's topological information. AS 65200, for example, has no way of knowing what the route through AS 65100 looks like; all it knows is that the destination is reachable through AS 65100. One interesting side effect of considering each autonomous system as a single entity for which the autonomous system route vector is connected is that BGP can only detect loops between autonomous systems without additional knowledge or laws, and it cannot guarantee loop-free paths within an AS as shown in Figure 8.

9.6 Types of Casting in a computer network

The cast term here signifies some data(stream of packets) is being transmitted to the recipient(s) from client(s) side over the communication channel that helps them to communicate. Let's see some of the "cast" concepts that are prevailing in the computer networks field.

1. Unicast

When only one sender and one receiver are involved, this method of data transfer is advantageous. So, in a nutshell, it is a one-to-one transmission. When a device with an IP address of 10.1.2.0 in one network needs to transfer traffic (data packets) to a device with an IP address of 20.12.4.2 in another network, unicast is used. Unicast example is shown in Figure 9.

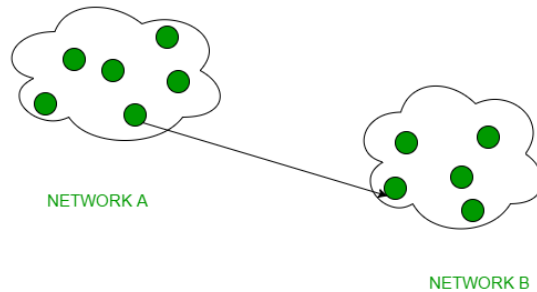


Figure 9 Unicast Example

Broadcast

There are two methods of broadcasting conversion (one-to-all) techniques:

Limited Broadcasting

If you need to transmit a stream of packets to all of the computers on your network, broadcasting will come in handy as shown in Figure 10. To do this, it will append 255.255.255 (all 32 bits of the IP address set to 1) to the destination address of the datagram (packet) header, which is reserved for information transmission to all recipients from a single client (sender) across the network.

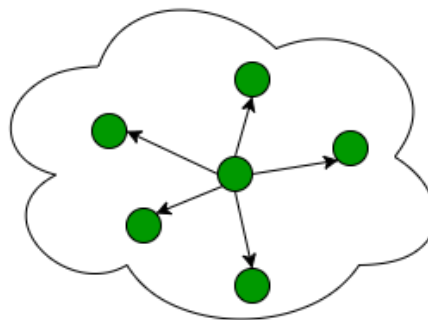


Figure 10 Network Cluster

Direct Broadcasting

If a node on one network needs to send a packet stream to all of the devices on the other network, this is helpful. This is accomplished by converting all of the destination address's Host ID portion bits to 1, which is referred to as Direct Broadcast Address in the datagram header for information transfer.

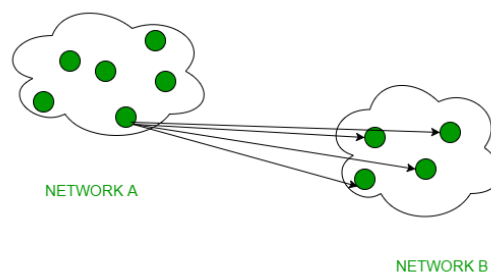


Figure 11 Direct Broadcasting

For video and audio delivery, television networks primarily use this mode. In computer networks, one of the most important protocols in this class is Address Resolution Protocol (ARP), which is used to resolve IP addresses into physical addresses, which is needed for underlying communication.

Multicast

In multicasting, data transfer traffic is shared between one or more senders and one or more receivers. The traffic in this system is split into unicast (one-to-one) and broadcast (multicast) (one-to-all). Multicast allows servers to send single copies of data streams to hosts that require them.

For IP multicast to operate, it needs the assistance of other protocols such as IGMP (Internet Group Management Protocol) and Multicast routing. Class D is also reserved for multicast classes in Classful IP addressing.

9.7 Routing in Adhoc Network

Challenges in Adhoc network

Dynamic topology, unreliable connections, restricted resources, lower connection bandwidth security, and no default router are all problems in an ad hoc network. There could be one other major problem in routing: there are no physical connections.

What does physical connection mean?

It means that when nodes travel from one location to another, wireless connections are formed and broken. As a result of the regular disconnections and partitions, no physical relations exist. As a result, routing protocols are classified as proactive, reactive, or hybrid as seen in Figure 12.

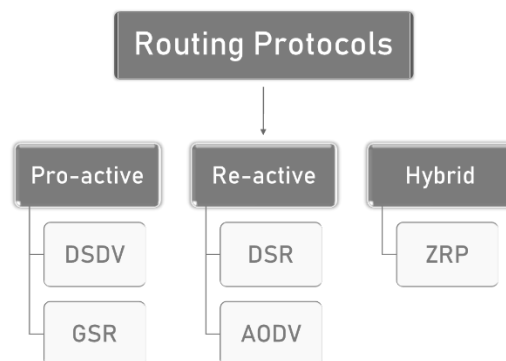


Figure 12 Routing Protocols

Proactive Routing protocols

First and foremost, in a constructive routing, proactive. Until packets are sent, the routing tables are established. The following is an example of a connection state distance factor. Each node in the network is aware of the routes to all other nodes. Each mobile node has its own routing table, which stores information about routes to all potential destination mobile nodes. So, in the case of constructive, because the topology in the mobile ad hoc network is complex, as I previously said, these routing tables are changed regularly, either every 30 seconds or 35 seconds, depending on how quickly the network topology varies. As a result, constructive routing protocol has a drawback. As a result, it does not operate well with large networks so the routing table entries become too large as they must retain and route information to all available nodes.

1. DSDV(Destination Sequenced Distance Vector Routing Protocol)

It is a proactive table driven routing protocol that extends the distance vector routing protocol and is built on the Bellman Ford routing algorithm. DSDV is a proactive table driven routing protocol that falls under the proactive. Due to the count to infinity issue, the distance vector routing protocol was not suitable for a mobile ad hoc network. So, we have a solution for this in the form of the destination sequence distance vector routing protocol. Any routing entry in the routing table that is managed by each node now has a destination sequence number associated with it. Only if the entry consists of a new modified path to the destination with a higher sequence number will a node contain the new upgrade in the table. As a result, it is proactive which is why it is referred to as a table guided routing protocol.

2. GSR(Global State Routing)

The next protocol is the global state routing protocol, which extends the wired network's connection state routing and is built on Dijkstra's routing algorithm. Since each node floods the link state routing information directly through the whole network, the link state routing protocol is not suitable for mobile ad hoc networks. As a result, global flooding can cause a network's control packets to become congested. In the last slides of the previous lecture, I explained distance relation state route vector routing. So, if you're not sure what's causing the rain, you should attend the session. What is connection state routing and how does it work? Since you can get all of the details from the floods, what is distance vector routing? As a result, the global state routing protocol was introduced as a workaround. Really, link state routing packets are not flooded into the network by global state routing.

Reactive

This are also known as on-demand routing protocols in the case of reactive routing. This path is only found when it is needed or required. Path exploration is accomplished by flooding route request packets across the mobile network. It is divided into two phases: path exploration and route management. The path is only discovered when it is required.

1. DSR(Dynamic Source Routing protocol)

The DSR dynamic source routing protocol is the next step. It's a reactive or on-demand routing protocol, which means the path is only found when it's required. Path exploration is accomplished by flooding route request packets across the mobile network. There are two stages of it: path exploration and route management.

Route Discovery

This process decides the most efficient route for data packet transfer between the source and destination mobile nodes.

Route Maintenance

Since the topology of a mobile ad-hoc network is complex, this step performs route maintenance. As a result, there are several instances of connection breakage resulting in network collapse between mobile nodes.

2. AODV (Ad-Hoc On Demand Vector Routing protocol)

It's an on-demand/reactive routing protocol. It is a dynamic source routing protocol (DSR) extension that aids in the elimination of the protocol's disadvantage. When the source mobile node sends the data packet to the destination mobile node after route discovery in DSR, it also includes the full path in its header. As a result, as the network grows in complexity, the length of the full route grows, as does the size of the data packet header, slowing down the entire network. As a result, the Ad-Hoc On Demand Vector Routing protocol was developed as a solution. The key distinction is about how the route is stored: AODV stores it in the routing table, while DSR stores it in the data packet's header. It works in the same way, with two phases: route exploration and route maintenance.

Hybrid routing Protocol

It essentially incorporates the benefits of both reactive and proactive routing protocols in one package. These protocols are flexible in design, adapting to the source and destination mobile nodes' zone and location. One of the most popular hybrid routing protocol is Zone Routing Protocol (ZRP).

Summary

- Whether the deliverer (host or router) and the destination are on the same network, the packet delivery is called direct; if the deliverer (host or router) and the destination are on separate networks, the packet delivery is called indirect.
- Instead of a full list of the stops the packet must make, the next-hop approach only lists the location of the next hop in the routing table; in the network specific method, all hosts on a network share one routing table entry.
- The full IP address of a host is given in the routing table in the host-specific method.

- To avoid massive routing tables, classless addressing necessitates hierarchical and regional routing.
- The entries in a static routing table are modified manually by an administrator, while the entries in a dynamic routing table are updated automatically by a routing protocol.
- RIP is based on distance vector routing, in which each router shares its knowledge of the entire AS with its neighbours at regular intervals.
- Dijkstra's algorithm is used to determine OSPF routing tables.
- BGP is a routing protocol for interautonomous systems that is used to update routing tables.
- Only packets that have travelled the shortest distance from the source to the router are forwarded in reverse path forwarding (RPF).
- Link State Routing tries to find its neighbours and learn their network addresses so that the router can choose the shortest path. Many classes are used to relay packets in hierarchical routing. Broadcast and multicast routings are used to send a single packet to multiple recipients based on whether they are members of a broadcast or multicast network.
- The shortest path to - destination in the network is identified by traversing the tree, and the Dijkstra algorithm is the most commonly used shortest path first algorithm.
- Based on the advertised information about the direction and distance for each destination, which are stored in a local database, the distance vector algorithms are used to decide which route is the safest path to each destination.

Keywords

Adaptive Algorithms: They can change their routing decisions in response to changes in the topology and traffic, and they can automatically adjust routing details as the network configuration changes.

Distance Vector Routing: It maintains a routing table and exchanges its routing table with each of its neighbors so that their routing tables get updated.

Flow-based Routing: It considers both the topology and the load.

Hierarchical routing: Intra-domain and inter-domain routing are used in hierarchical routing.

Link State Routing allows each router in the network to learn the network topology and build a routing table based on it.

Multicast is a term that refers to one or more network interfaces that are spread across several subnets. One-to-many connectivity is possible.

Multicast Routing: This refers to transmitting data to well-defined groups with a large number of participants but a limited number in comparison to the whole network.

Review Questions

1. Describe briefly how hierarchal algorithm works.
2. What is the main purpose of using router in a network?
3. Differentiate between:
 - a. Connectionless and connection-oriented service
 - b. Interior and Exterior Routing
 - c. Link state and distance vector routing
4. Difference between proactive and reactive routing protocols.
5. What is the purpose of BGP?
6. Why do OSPF messages propagate faster than RIP messages?

Self-Assessment

1. Multiple access schemes are used to allow _____ mobile users to share simultaneously a finite amount of radio spectrum.
 - a. Many
 - b. One
 - c. Two
 - d. Ten-Fifteen
2. Multiple access protocols is divided into
 - a. One
 - b. Two
 - c. Three
 - d. Four
3. The use of hierarchy in routing tables can _____ the size of routing tables.
 - a. Reduce
 - b. Increase
 - c. Both a and b
 - d. None of the above.
4. If there is only one routing sequence for each source destination pair, the scheme is known as
 - a. static routing
 - b. fixed alternative routing
 - c. standard routing
 - d. dynamic routing
5. Count-to-Infinity problem occurs in
 - a. distance vector routing
 - b. short path first
 - c. link state routing
 - d. hierarchical routing
6. Link state packets are built in
 - a. short path first
 - b. distance vector routing
 - c. link state routing
 - d. hierarchical routing
7. In which routing method do all the routers have a common database?
 - a. Distance Vector
 - b. Link Vector
 - c. Shortest path
 - d. Link State
8. In distance vector routing algorithm, the routing tables are updated
 - a. by exchanging information with the neighbours
 - b. automatically
 - c. using the backup database
 - d. by the server
9. Distance vector routing algorithm is implemented in Internet as

- a. OSPF
 - b. RIP
 - c. ARP
 - d. APR
10. Which of the following routing algorithm takes into account the current network load.
- a. broadcast
 - b. shortest path
 - c. flooding
 - d. distance vector routing
11. In distance vector routing the delay metric is
- a. number of hops
 - b. geographical distance
 - c. number of neighbours
 - d. queue length
12. A well-defined groups that are numerically large in size but small compared to the network as a whole are used in
- a. Unicast routing
 - b. Multicast routing
 - c. Broadcast routing
 - d. Telecast routing

Answers

- | | | | | | | | |
|----|---|-----|---|-----|---|-----|---|
| 1. | a | 2. | c | 3. | a | 4. | b |
| 5. | a | 6. | b | 7. | d | 8. | a |
| 9. | b | 10. | d | 11. | d | 12. | b |

Further Readings



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